Government Engineering College
Bhavnagar

Programming Assignment

Subject Name : Microprocessor and Interfacing
Semester-Branch : 6th CE
Subject Code : 3160712
Academic Year : 2025-26
Subject in-charge : Prof. H. S. Sanghavi
Submission Date : 16-04-2026

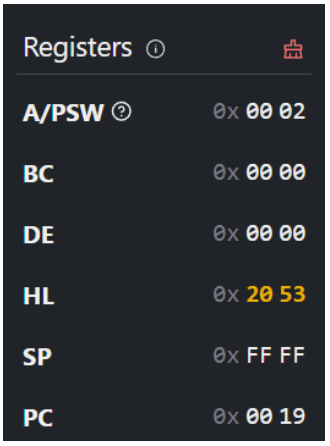
1) Write an 8085 Assembly language program to interchange 16-bit data stored in memory locations 2050, 2051, 2052, and 2053. WITHOUT USING XCHG INSTRUCTION.

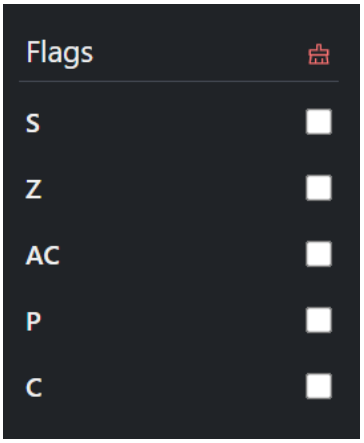
Definition:
This program is used to swap two 16-bit numbers stored in memory locations 2050H–2051H and 2052H–2053H without using the XCHG instruction, by transferring data through registers and then storing it back in exchanged positions.

8085 Code:

```
LXI H,2050H     ; HL → first number
MOV A,M         ; A = low byte (1st)
INX H
MOV B,M         ; B = high byte (1st)

LXI H,2052H     ; HL → second number
MOV C,M         ; C = low byte (2nd)
INX H
MOV D,M         ; D = high byte (2nd)

; Store 2nd → 1st
LXI H,2050H
MOV M,C
INX H
MOV M,D

; Store 1st → 2nd
LXI H,2052H
MOV M,A
INX H
MOV M,B

HLT
```

OUTPUT





2) The memory location 2070H holds the data byte F2H. Write instructions to transfer the data byte to the accumulator using three different Opcode: MOV, LDAX, and LDA.

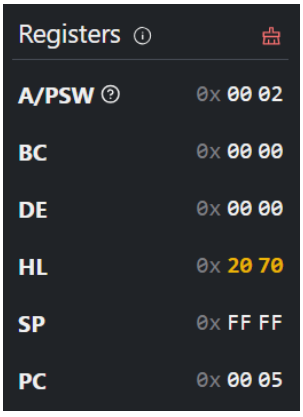
Definition:

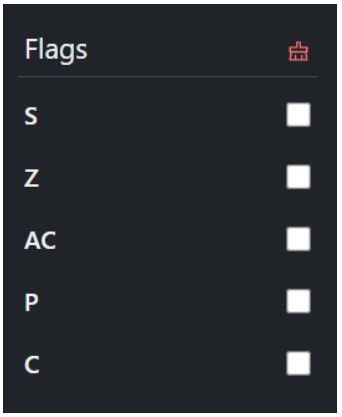
This program is used to load a data byte from memory location 2070H into the accumulator using three different instructions—MOV, LDAX, and LDA—each of which accesses memory in a different manner.

8085 Code:

(a) LXI H,2070H
MOV A,M
HLT

OUTPUT:

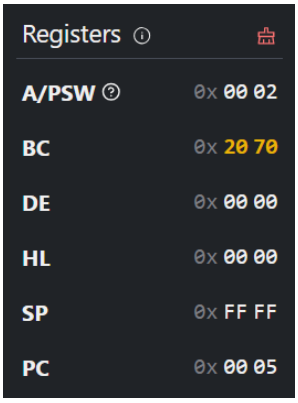


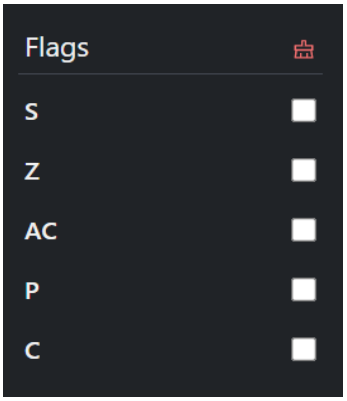


8085 Code:

(b) LXI B,2070H
LDAX B
HLT

OUTPUT:



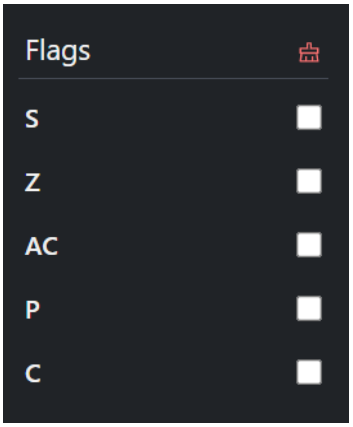


8085 Code:

(c) Using LDA

```
LDA 2070H
HLT
```

OUTPUT:



3) Write assembly language program to do multiplication of two numbers.

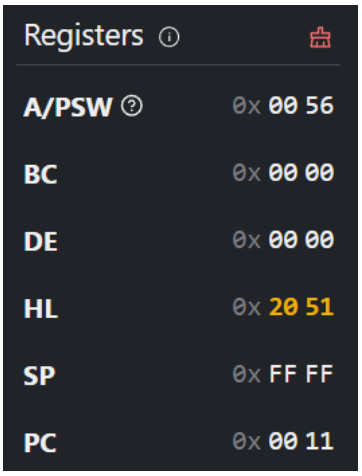
Definition:

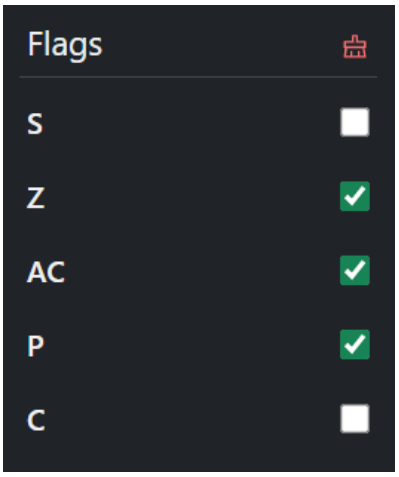
This program is used to multiply two 8-bit numbers stored in memory. The multiplication is performed using repeated addition method. The final result is stored in a specified memory location.

8085 Code:

```
LXI H,2050H
MOV B,M      ; multiplicand
INX H
MOV C,M      ; multiplier

MVI A,00H    ; result = 0

LOOP: ADD B
      DCR C
      JNZ LOOP

STA 2052H    ; store result
HLT
```

OUTPUT:





4) Write a set of 8085 assembly language instructions to generate a 0.5 second delay, if the crystal frequency is 4 MHz.

Definition:

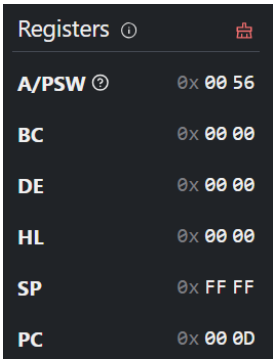
This program is used to generate a time delay of approximately 0.5 seconds. It uses nested loops and instruction execution time based on a 4 MHz clock frequency. The delay is achieved by continuously decrementing register values.

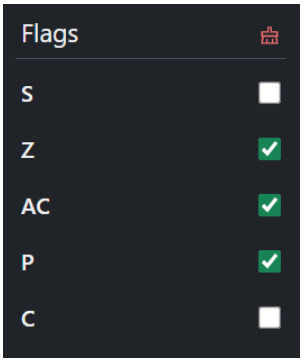
8085 Code:

```
MVI B,FFH

LOOP1: MVI C,FFH
LOOP2: DCR C
       JNZ LOOP2
       DCR B
       JNZ LOOP1

HLT
```

Note: Loop values are reduced for simulation purpose. Actual delay uses FFH for required timing.

OUTPUT:





5) Ten 8-bit values are stored from memory location 5000H onwards. Write an 8085-assembly language program to add POSITIVE values on addresses starts from 5100H and NEGATIVE values on addresses starts from 5200H.

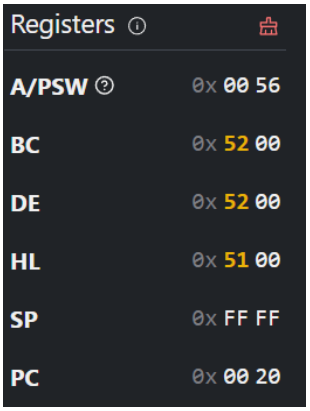
Definition:

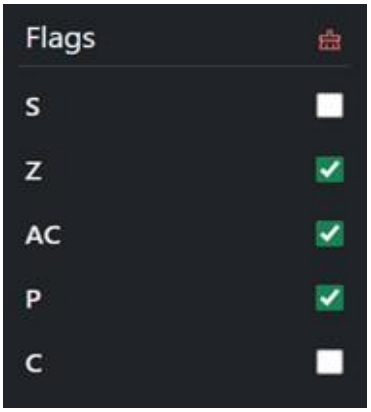
This program is used to separate positive and negative numbers from a given array of data. It checks the sign bit of each number to determine whether it is positive or negative. The positive numbers are stored at memory locations starting from 5100H and negative numbers from 5200H.

8085 Code:

```
    LXI H,5000H
    MVI C,0AH

    LXI D,5100H     ; positive storage
    LXI B,5200H     ; negative storage

    LOOP: MOV A,M
        ANI 80H

        JZ POSITIVE

    NEG:  MOV A,M
        STAX B
        INX B
        JMP NEXT

    POSITIVE:
        MOV A,M
        STAX D
        INX D

    NEXT: INX H
        DCR C
        JNZ LOOP
    HLT
```

OUTPUT:





6) Write an 8085 assembly language program to arrange the following numbers in ascending order: 29H, 47H, 06H, 03H, 17H.

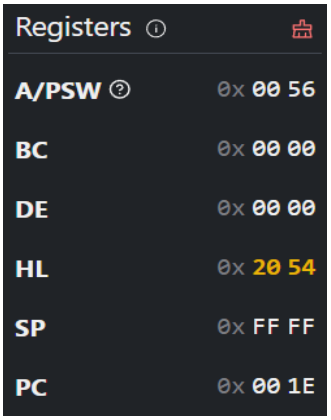
Definition:

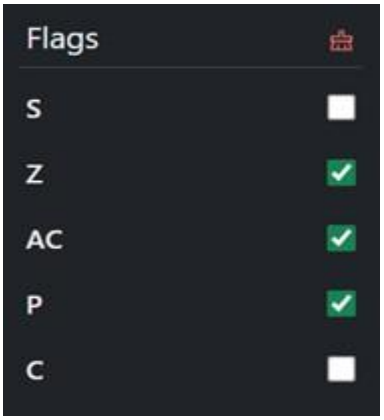
This program is used to arrange a set of numbers in ascending order. It uses the bubble sort technique by comparing adjacent numbers and swapping them if required. The sorted result is stored in the same memory locations.

8085 Code:

```
LXI H,2050H
MVI C,04H

OUTER: LXI H,2050H
       MVI B,04H

INNER: MOV A,M
       INX H
       CMP M
       JC SKIP

       MOV D,M
       MOV M,A
       DCX H
       MOV M,D
       INX H

SKIP:  DCR B
       JNZ INNER

       DCR C
       JNZ OUTER

HLT
```

OUTPUT:





7) An array of ten numbers is stored from memory location 2000H onwards. Write an 8085 assembly language program to separate out and store the EVEN and ODD numbers on new arrays from 2100H and 2200H, respectively

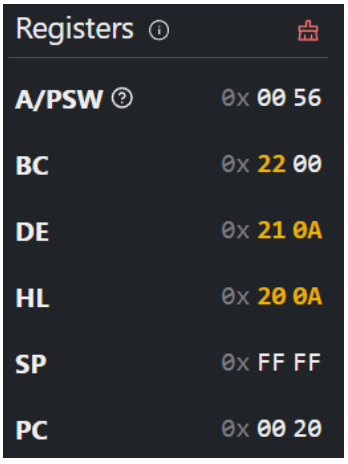
Definition:

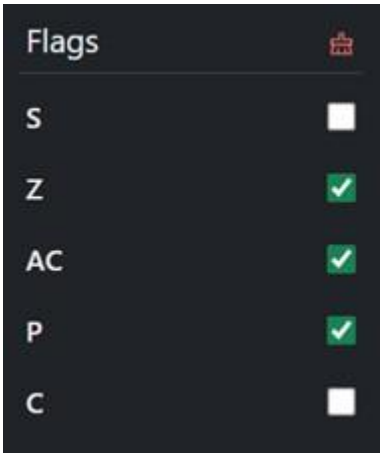
This program is used to separate even and odd numbers from a given array. It checks the least significant bit of each number to determine whether it is even or odd. The even numbers are stored starting from 2100H and odd numbers from 2200H.

8085 Code:

```
LXI H,2000H
LXI D,2100H     ; EVEN
LXI B,2200H     ; ODD

MVI C,0AH

LOOP: MOV A,M
    ANI 01H
    JZ EVEN

ODD:  MOV A,M
    STAX B
    INX B
    JMP NEXT

EVEN: MOV A,M
    STAX D
    INX D
NEXT: INX H
    DCR C
    JNZ LOOP

HLT
```

OUTPUT:





8) Write an 8085 assembly language program to count the number of bytes that are greater than 2010 and lesser than 4010 from an array of ten bytes stored on memory locations 2000H onwards. Store such numbers on memory locations 3000H onwards.

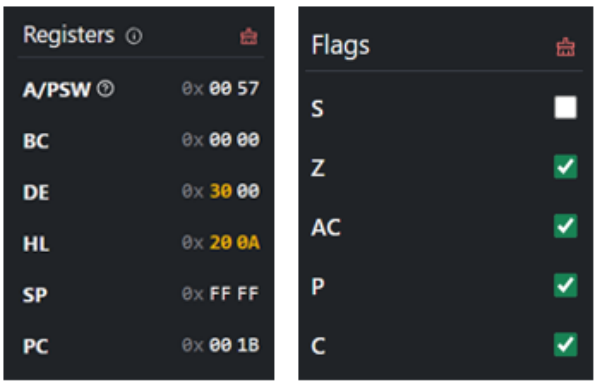
Definition:

This program is used to identify numbers within a specific range from an array. It checks whether each number is greater than 20H and less than 40H. The numbers satisfying the condition are stored in memory locations starting from 3000H.

8085 Code:

```
LXI H,2000H     ; HL → source memory (2000H)
LXI D,3000H     ; DE → destination memory (3000H)

MVI C,0AH       ; Counter = 10 bytes

LOOP: MOV A,M   ; Load data from memory into A

CPI 21H         ; Compare with 21H
JC SKIP         ; If A < 21H → skip

CPI 40H         ; Compare with 40H
JNC SKIP        ; If A ≥ 40H → skip

STAX D          ; Store valid data at DE
INX D           ; Increment destination pointer

SKIP: INX H     ; Move to next source location
DCR C           ; Decrement counter
JNZ LOOP        ; Repeat until C = 0

HLT             ; Stop
```

OUTPUT:



9) The following block of data is stored in the memory locations from 2050H to 2055H. Write an 8085 Assembly language program to Transfer the data to the locations 2080H to 2085H in the reverse order. DATA(H) 25, A5, 4F, E3, AF, F1

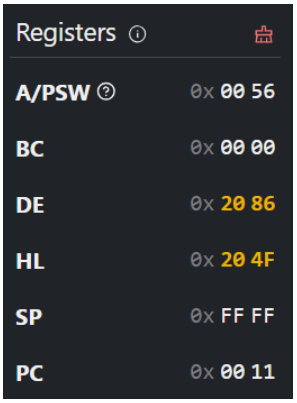
Definition:

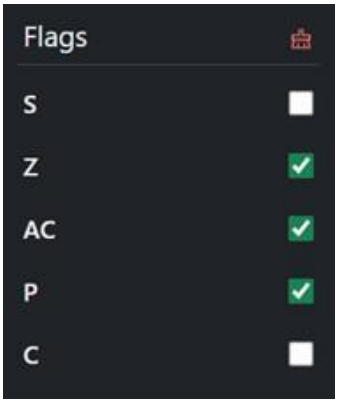
This program is used to transfer a block of data from one memory location to another in reverse order. The source data is stored from 2050H to 2055H. The reversed data is stored from 2080H onwards.

8085 Code:

```
LXI H,2055H     ; HL → last element of source
LXI D,2080H     ; DE → destination start

MVI C,06H       ; Counter = 6 elements

LOOP: MOV A,M   ; Get data
    STAX D      ; Store at destination
    DCX H       ; Move backward in source
    INX D       ; Move forward in destination

    DCR C
    JNZ LOOP

HLT
```

OUTPUT:





10) An array of ten data bytes is stored on memory locations 2100H onwards. Write an 8085 assembly language program to find the largest number and store it on memory location 2200H.

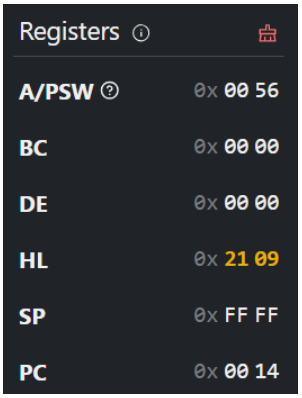
Definition:

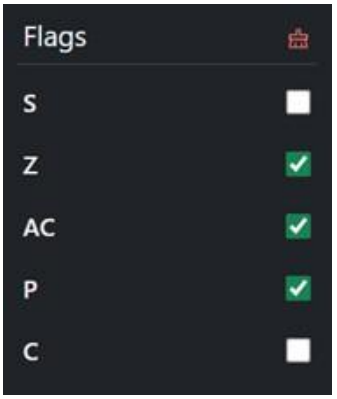
This program is used to find the largest number from a given array of data. It compares each number with the current maximum value. The largest number is stored at memory location 2200H.

8085 Code:

```
LXI H,2100H     ; HL → start of array
MOV A,M         ; Assume first number is largest
MVI C,09H       ; Remaining 9 numbers
LOOP: INX H     ; Move to next element
    CMP M       ; Compare A with memory
    JNC SKIP    ; If A ≥ M → skip
    MOV A,M     ; Update largest
SKIP: DCR C
    JNZ LOOP
STA 2200H       ; Store result at 2200H
HLT
```

OUTPUT:



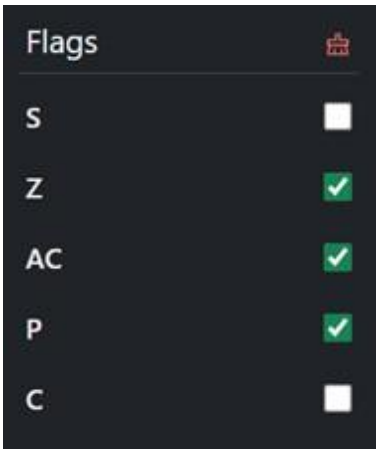


11) Write an 8085 Assembly language program to evaluate the Boolean equation D = (B+C) * E, where B, C, E represents data in various registers of 8085.

Definition:

This program is used to compute the expression D = (B + C) × E by first adding the values in registers B and C, then multiplying the result with register E using repeated addition, and finally storing the result in register D.

8085 Code:

```
MOV A,B
ADD C        ; A = B + C

MOV H,A      ; H = (B+C)

MVI D,00H    ; result = 0

LOOP: MOV A,D
      ADD H
      MOV D,A

      DCR E
      JNZ LOOP

HLT
```

OUTPUT: